

Note on naming: The uploaded 2025 publication is titled “Land Cover Atlas of Afghanistan 2024” and was published by FAO in Rome in 2025 (DOI: 10.4060/cd3186en). In this note it is referred to as the “2025 FAO Land Cover Updated Atlas,” reflecting the publication year of the updated product.

1. Methodological details of the 2025 updated FAO Land Cover Atlas

1.1 Purpose and institutional framework

The 2025 FAO Land Cover Atlas updates Afghanistan's previous national land-cover database and atlas (FAO 2012/2016). It provides a high-resolution, nationally consistent land-cover dataset structured at national, provincial, district and watershed-basin levels to support food security planning, agricultural monitoring, natural-resources management, disaster preparedness and evidence-based policy. The publication acknowledges EU assistance and collaboration with national experts and institutions, including use of NSIA administrative datasets.

1.2 Overall methodological workflow

The Atlas follows a three-phase production chain:

Phase 1 – Preliminary analysis: study of the Area of Interest (AOI) using Afghanistan's 18 agro-ecological zones; preparation of the land-cover legend and photo-keys; selection of ancillary datasets; definition of the cloud-computing environment.

Phase 2 – Model development: imagery selection/acquisition/processing; feature enrichment (spectral indices, texture and statistics); training-data collection; automated irrigated vs rainfed discrimination; Random Forest model definition; hierarchical land-cover classification.

Phase 3 – Results generation: accuracy assessment; map harmonization; zonal statistics; comparative assessment versus the previous atlas. Model development and results generation were iterated until convergence.

1.3 Technology and classification approach

Technology stack and analytical approach include:

- Cloud-based geospatial computing on Google Earth Engine (GEE), enabling national-scale processing without downloading the full raster data cube locally.
- Machine learning: Random Forest classifier implemented in GEE via the Python API (ensemble decision-tree model with bootstrap sampling).
- Hierarchical land-cover classification producing a detailed map of 21–24 classes, later aggregated to 11 communication classes for atlas presentation.
- Automated irrigated/rainfed separation tool combining CHIRPS precipitation and FAO WaPOR actual evapotranspiration (ETa): irrigated where ETa exceeds precipitation; rainfed where precipitation meets/exceeds ETa; also used to infer cropping-cycle intensity.
- Multi-sensor / multi-source feature stack: Sentinel-2 optical time series + DEM topography + precipitation + ETa.
- Training/validation supported by photointerpretation on Google Earth Pro and Collect Earth Online (CEO), previous land-cover references (including ICIMOD), and field ground-truth.
- Area estimation adjusted using confusion-matrix information (user's accuracy / commission redistribution) with 95% confidence intervals.

1.4 Software and digital tools

- Google Earth Engine (GEE) – primary cloud platform for data management, model development and map generation (JavaScript and Python interfaces).
- Python API for GEE – Random Forest training/prediction and workflow automation.
- Google Earth Pro – review and confirmation of sample points.
- Collect Earth Online (CEO) – structured high-resolution visual interpretation of reference samples.
- UN Geospatial / NSIA boundary datasets – cartographic base and statistical units.
- Publication production: Adobe InDesign (metadata of the uploaded PDF indicates InDesign 21.2).
- Outputs hosted also via FAO Agro-informatics Platform (atlas appendix reference).

1.5 Satellite / geospatial data inputs and resolution granularity

Primary remote-sensing and ancillary inputs:

Dataset / layer	Role	Spatial resolution / notes
Sentinel-2 Surface Reflectance (S2A/S2B)	Core optical classifier input (multispectral, multi-temporal)	10–60 m native bands; key analysis bands used after correlation screening; cloud probability masking; monthly/seasonal median composites Oct 2023–Sep 2024
Digital Elevation Model (USGS DEM)	Topographic context for class discrimination	30 m
CHIRPS precipitation	Irrigated vs rainfed discrimination	~5,566 m; Gaussian low-pass smoothing applied
FAO WaPOR actual evapotranspiration (ETa)	Irrigated vs rainfed / crop-cycle support	300 m; Gaussian smoothing applied
High-resolution basemaps (Google Earth / CEO) + field samples	Training, validation, photo-keys	Sub-metre to metre visual reference; not the national mapping pixel grid

Effective mapping granularity: national land-cover mapping is driven by Sentinel-2 at approximately 10 m for key spectral bands, which is a major upgrade relative to the 2016 atlas SPOT-4 10 m object-based polygons and Landsat fill-ins, and especially relative to five-year “potential” agricultural estimates in earlier assessments. The Atlas explicitly characterizes the 2024 product as a single-year, observation-based annual snapshot.

1.6 Land-cover legend and standards

The legend is based on FAO’s Land Cover Meta Language (LCML; ISO 2012/2023), itself derived from the Land Cover Classification System (LCCS; Di Gregorio & Jansen, 2000). The Atlas defines 24 detailed subclasses grouped into 11 aggregated classes for statistics and map reading:

Built-up; Fruit trees; Vineyards; Irrigated herbaceous cropland; Rainfed agricultural land; Forest and shrubland; Rangeland; Bare areas; Sand cover; Waterbodies and marshland; Snow cover.

Irrigated agricultural land (for trend comparison) is defined as the sum of irrigated herbaceous cropland + fruit trees + vineyards. Rainfed land includes flat and sloping rainfed subclasses. Hierarchical modelling first treats some agricultural subclasses in aggregated form for robustness, then refines where needed.

1.7 Training samples, accuracy and area estimation

- Training samples collected iteratively with stratified/adaptive refinement; poorly performing classes received additional points.
- Accuracy assessment used two-level stratified sampling: 100 × 100 km national grid, then stratified random split with ~25% held out for testing.
- 26,115 validation points used for the initial 21-class aggregated agricultural classification.
- Overall Accuracy = 0.872 (87.2%); Kappa = 0.858.
- Per-class UA/PA/F1 reported; common confusions noted among spectrally similar vegetated and bare classes.
- Model-driven area estimation adjusted with confusion-matrix proportions and 95% confidence intervals.

1.8 Geographic reporting units and total land area

Reporting is provided for Afghanistan's 34 provinces, 422 districts, 8 regions, and 5 major hydrological river basins (Harirod-Murghab; Northern; Panj-Amu; Kabul; Helmand) with 36 watersheds. For land-area calculations the Atlas adopts NSIA 2024 figures; national totals in the provincial summary sum to 64,442,023 ha (~64.44 million ha). Separately, the abstract cites 652,867 km² as the FAOSTAT figure last reported by the national statistical office, noting slight variations by source/method (NSIA 2024, NSIA 2025).

2. Methodological comparison: 2016 FAO Atlas, NSIA subsequent updates, and 2025 FAO Updated Atlas

The table below compares key methodological aspects. "NSIA subsequent updates" refers to post-2016 NSIA remote-sensing/satellite-interpretation products used for cultivated-area and related agricultural statistics (and NSIA administrative boundary datasets), which are not a full multi-class national land-cover atlas in the same sense as the FAO publications.

Methodological aspect	2016 FAO Land Cover Atlas(based on ~2010 database; published 2016)	NSIA subsequent updates(post-2016 operational RS products)	2025 FAO Land Cover Updated Atlas(Land Cover Atlas of Afghanistan 2024; published 2025)
Primary purpose	National land-cover database/atlas for food security, agricultural statistics and NRM; enrichment of 2010 database under TCP/AFG/3501	Operational estimation of cultivated/arable land, crop area (e.g. wheat) and yields; administrative statistics support	Updated national multi-class land-cover atlas for policy, food security, climate/adaptation and NRM; national–provincial–district–basin statistics
Core technology	FAO/GLCN object-based remote sensing + visual interpretation/enrichment with VHR imagery/aerial photos	Remote sensing and satellite-image interpretation for cultivation/crop monitoring (as publicly reported by NSIA)	Cloud computing + machine learning (Random Forest) on multi-temporal Sentinel-2; hierarchical classification; automated irrigated/rainfed tool (CHIRPS + WaPOR)

Main software / platforms	FAO MADCAT (Mapping Device Change Analysis Tools) / Terra Nova GeoVis suite; LCCS tools	Not fully documented in public technical atlas form; satellite interpretation / RS workflows; NSIA statistical systems	Google Earth Engine (Python/JavaScript); Google Earth Pro; Collect Earth Online; FAO Agro-informatics dissemination; InDesign for atlas layout
Primary satellite / imagery sources	SPOT-4 (mainly 2009–2011; ~240 scenes; ~90% coverage), Landsat GLS-2011 fill-in; VHR aerial photos / HR satellite; Google Earth support	Satellite imagery (sensor mix not always specified in public press releases); annual/seasonal cultivation assessments	Sentinel-2 SR (Oct 2023–Sep 2024), USGS DEM, CHIRPS precipitation, FAO WaPOR ETa; HR basemaps for labelling
Spatial resolution / mapping granularity	SPOT multispectral merged products at ~10 m for mapping; Landsat 30 m fill; VHR ~0.5–1 m for enrichment; database scale ~1:50,000; >500,000 polygons	Operational cultivated-land mapping; typically coarser thematic focus (cultivated vs irrigated/rainfed crop extents) rather than full 11-class atlas polygons	Sentinel-2-driven ~10 m effective mapping for key bands; DEM 30 m; auxiliary climate layers at 300 m–5.5 km smoothed for irrigated/rainfed assistance
Classification system / legend standard	FAO LCCS; 25 detailed classes aggregated to 11 generalized classes (URB, AGT, AGV, AGI, AGR, NFS, NHS, BRS, BSD, WAT, SNW)	Focus on agricultural/cultivated classes (irrigated vs rainfed, crop-specific extents); not a full LCCS atlas republished as FAO-style product	FAO LCML/LCCS lineage; 24 subclasses → 11 aggregated classes; UML/LCML diagrams and photo-keys in atlas
Classification method	Object-based segmentation + interactive/visual labelling in MADCAT; enrichment/editing with aerial photos and VHR	Image interpretation / RS-based cultivated-area delineation (methodological detail less formally published than FAO atlases)	Pixel-/sample-based supervised Random Forest; hierarchical class structure; iterative sample enrichment
Temporal basis	Multi-year imagery window (~2009–2011) representing circa-2010 land cover; atlas published 2016; earlier assessments described as multi-year/potential agricultural estimates	Annual or campaign-based cultivated-area updates (solar-year reporting)	Single-year observation-based snapshot (2023/24 growing-year composites)
Training / reference data	Visual interpretation, aerial photos, ancillary topographic/irrigation layers; field constraints due to security	Operational RS interpretation; administrative/agricultural reporting needs	Tens of thousands of points; Google Earth Pro + CEO; prior maps; field ground-truth; iterative adaptive sampling

Accuracy assessment	Quality control via VHR enrichment and expert interpretation; classical atlas QC (formal OA/Kappa less emphasized in summary materials than 2025)	Not consistently published as OA/Kappa for a full multi-class land-cover map	Formal accuracy assessment: OA 87.2%, Kappa 0.858 on 26,115 test points; UA/PA/F1 and confusion matrices published
Cloud / computing architecture	Desktop/enterprise GIS + FAO MADCAT workflows; local mosaics/segmentation	National statistical RS production environment (details not fully public)	Fully cloud-native GEE workflow for data cube handling and classification
Administrative / statistical units	34 provinces and districts; 5 basins; attributes for PROV_NAME / DIST_NAME from counterpart agency layers	NSIA official admin units; cultivated statistics by province	34 provinces, 422 districts, 8 regions, 5 basins / 36 watersheds; NSIA admin boundaries + UN Geodata for international base
Coordinate reference / distribution	Shapefile / geodatabase; UTM Zone 42N, WGS-84	National statistical products / reports	National land-cover rasters/maps + zonal tables; cloud/platform dissemination
Key methodological advance vs previous	Upgrade from earlier 1990s Landsat-era products; SPOT 10 m OBIA + LCCS standardization	More frequent cultivated-area monitoring than static atlas cycles	Shift from multi-year potential/object atlas to ML + Sentinel-2 single-year mapping; automated irrigated/rainfed separation; quantified accuracy and CI-based area estimates

Important caveat: Because NSIA updates emphasize cultivated/crop statistics rather than a complete FAO-style multi-class land-cover atlas, several rows necessarily describe NSIA's operational RS cultivation monitoring rather than an equivalent 11-class national land-cover legend.

3. UN cartographic rules / procedures adhered to in the Atlas

The 2025 FAO Atlas explicitly aligns sensitive geographic elements with United Nations geospatial practice and FAO publication standards. The key practices relevant to the user's questions are summarized below.

3.1 Standard UN / FAO boundary disclaimer

The Atlas includes the standard United Nations Secretariat disclaimer (also used across FAO map products):

"The designations employed and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the Secretariat of the United Nations concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries."

Map pages repeatedly instruct readers to "Refer to the disclaimer on page ii for the names and boundaries used in this map," ensuring the disclaimer applies to all national/provincial map plates.

3.2 Demarcation of the international boundary

International boundaries and base geographic layers were prepared using UN Geodata / UN Geospatial (2020) to ensure alignment with widely recognized global standards. In map legends, “International boundary” is symbolized distinctly from provincial boundaries. Neighbouring countries are described in geographic narrative (Pakistan to the east/south; Islamic Republic of Iran to the west; Turkmenistan, Uzbekistan and Tajikistan to the north; China to the northeast via the Wakhan Corridor), while the legal disclaimer prevents the map from being read as a sovereignty determination.

3.3 Marking of contested / disputed international boundaries

The Atlas legend includes a specific symbol for “Disputed boundary.” This is consistent with UN cartographic practice of differentiating undisputed international boundaries from contested lines. In the Afghanistan regional context, UN maps commonly also apply the standard Jammu & Kashmir note (dotted Line of Control; final status not agreed by the parties). The Atlas narrative refers to the border with the Jammu and Kashmir region “as currently recognized in agreements between India and Pakistan,” which mirrors the UN’s cautious, status-neutral formulation. FAO’s broader map-disclaimer framework likewise includes the dotted-line LoC language for Jammu and Kashmir.

Operationally, this means contested segments are not drawn as ordinary solid international boundaries without qualification; they are marked with disputed-boundary symbology and covered by the UN/FAO non-endorsement disclaimer.

3.4 Total land area of the country

UN/FAO practice separates (a) cartographic depiction of boundaries from (b) the statistical land-area figure used for national accounting. The Atlas follows that separation explicitly:

- Cartographic international base: UN Geospatial / UN Geodata.
- Land-area calculation for atlas statistics: NSIA 2024 dataset, “to maintain consistency with officially agreed-upon national statistics while preserving compatibility with internationally endorsed boundary representations.”
- Reported statistical total in provincial land-cover summary: 64,442,023 ha (~64.44 million ha).
- Abstract also cites 652,867 km² as the figure last reported by the national statistical office to FAOSTAT, noting slight source/method variations (NSIA 2024, NSIA 2025).

This dual approach is a practical UN-compatible procedure: use internationally endorsed boundary geometry for map portrayal, while adopting nationally agreed area totals for statistical reporting where those are the authoritative domestic figures.

3.5 Sub-national / provincial and district administrative boundaries

Sub-national boundaries are taken from NSIA administrative shapefiles (personal communication / unpublished NSIA dataset, 2024), combined with UN geospatial references for consistency. The Atlas analyzes and reports land cover across:

- 8 regions
- 34 provinces
- 422 districts
- 5 river basins / 36 watersheds (hydrological units from AIMS/HDX sources, used for basin statistics rather than political boundaries)

Map plates distinguish national capital, provincial capitals, international boundaries, provincial boundaries and disputed boundaries. Because administrative names/limits can change and may be sensitive, repeated referral to the page-ii disclaimer is used—again standard UN cartographic risk management.

3.6 Additional UN/FAO cartographic good practices reflected in the Atlas

- Neutrality of designations: map presentation does not express opinion on legal status of territories or authorities.

- Source citation on map plates: NSIA (2024) and UN Geospatial (2020) are cited under figures.
- Clear legend hierarchy: separate symbols for international, provincial and disputed boundaries.
- Multi-scale statistical presentation without changing the underlying international base geometry.
- Licence and third-party material notices (CC BY-NC-SA), plus statement that views are those of authors and not necessarily FAO policy.

4. Comparison of major land-cover areas: 2016 FAO Atlas, NSIA updates, and 2025 FAO Updated Atlas

Figures for the 2016 FAO Atlas are taken from the published national land-cover legend/hectarage tables (circa-2010 database presented in the 2016 atlas). Figures for the 2025 FAO Updated Atlas are taken from the atlas abstract, Table 6 (2016 vs 2024 comparative assessment), and the national provincial summary (Table 7). NSIA does not publish an equivalent full 11-class national land-cover atlas on the same schedule; therefore NSIA columns report the closest publicly available cultivated-land statistics and note “n/a” where a comparable national class area is not published.

Land classification / type	2016 FAO Land Cover Atlas(national hectarage / %)	NSIA subsequent updates(publicly reported cultivated-land figures; not a full LC atlas)	2025 FAO Land Cover Updated Atlas(2024 assessment; million ha / %)	Observed change 2016 FAO → 2025 FAO (as reported in Atlas Table 6 / text)
Total country land area used for statistics	About 64.4–65.2 million ha depending on table rounding / national frame (~652,000 km² cited in 2016 materials)	NSIA/FAOSTAT national area references vary slightly by year/method; Atlas cites NSIA 2024/2025 variation around FAOSTAT 652,867 km²	64.442 million ha (64,442,023 ha) in provincial summary; abstract also cites 652,867 km² FAOSTAT figure	Statistical frame aligned to NSIA 2024 for atlas area calculation; compare class shares with caution
Built-up (urban + non-urban)	306,855 ha (0.48%)— Urban 280,478 ha; Non-urban 26,377 ha	No full national built-up class series equivalent to FAO atlas publicly released as LC atlas update	0.7 million ha (1.0%)— Table 7: 664,747 ha	Increase; nearly doubled (0.31 → 0.7 million ha; 0.5% → 1.0%)
Fruit trees (orchards)	117,642 ha (0.18%)	Not published as standalone national LC class in NSIA cultivation bulletins	0.4 million ha (0.6%)— Table 7: 408,444 ha	Marked increase (0.12 → 0.4 million ha)
Vineyards	82,450 ha (0.13%)	Not published as standalone national LC class in NSIA cultivation bulletins	0.1 million ha (0.2%)— Table 7: 118,903 ha	Slight/moderate increase (0.08 → 0.1 million ha)

Irrigated herbaceous cropland(annual irrigated crops)	Intensively cultivated components in 2016 legend:3a 2,490,480 ha; 3a1 349,618 ha; 3c Active Karez 253,756 ha(+ marginal/poorly irrigated 3b 1,109,730 ha treated separately in 2016)	NSIA reports irrigated farmland within cultivated land, e.g. ~2.1 million ha irrigated among >3 million ha cultivated in recent public statements; wheat irrigated area examples ~1.33 million ha in a recent solar-year wheat assessment (crop-specific, not full irrigated LC)	3.5 million ha (5.4%)— Table 7: 3,484,287 ha	Slight decrease in irrigated herbaceous cropland (–115,923 ha; 3.6 → 3.5 million ha)
Total irrigated agricultural land(herbaceous + fruit trees + vineyards; FAO 2025 definition)	Approx. 5.9% of national area in comparative Table 6 baseline(2016 irrigated agricultural complex; class structure differs because 2016 also separated marginal/karez classes)	Public NSIA figures focus on irrigated cultivated land (~2.1 million ha in some recent cultivated-land statements)—method/definition not identical to FAO LC irrigated aggregation	6.2% of national area(sum of irrigated herbaceous + fruit trees + vineyards ≈ 4.01 million ha using Table 7)	Slight increase overall irrigated land (5.9% → 6.2%), driven mainly by permanent irrigated crops (fruit/vine) despite herbaceous decline
Rainfed agricultural land	3,734,494 ha (5.80%)— Flat 906,273 ha; Sloping 2,828,221 ha	NSIA rainfed cultivated land in recent public figures often cited around ~0.87–0.92 million ha within total cultivated land >3 million ha (definition = currently cultivated rainfed, not full rainfed LC mask)	3.1 million ha (4.8%)— Table 7: 3,077,928 ha	Decrease (–656,566 ha; 3.7 → 3.1 million ha; 5.8% → 4.8%)
Forest and shrubland	Needle-leaved forests + undifferentiated trees + high shrubs ≈ 1.78 million ha combined in detailed legend (~2.8% in comparative table)	No equivalent full forest/shrub national LC update series in NSIA cultivation releases	1.5 million ha (2.1–2.2%)— Table 7: 1,446,196 ha (2.2%)	Slight decrease (1.7 → 1.5 million ha)
Rangeland	30,243,985 ha (46.97%)	Not a primary NSIA cultivation-statistics class	28.6 million ha (44.4%)— Table 7: 28,587,735 ha	Decrease (–~1.7 million ha; 47% → 44.4%)

Bare areas (soil/rock)	Bare soil/rock outcrops 17,404,540 ha (27.03%) within broader bare/sand complex	n/a as FAO-equivalent class in public NSIA cultivation updates	18.5 million ha (28.6–28.7%)— Table 7: 18,446,153 ha (28.6%)	Increase (17.4 → 18.5 million ha)
Sand cover (sand + dunes)	Sand covered areas + dunes ≈ 4.78 million ha (7.4%) in comparative table	n/a	6.5 million ha (10.1%)— Table 7: 6,484,888 ha	Increase (4.78 → 6.5 million ha); Atlas abstract notes bare/sand covered increase ~4.3 percentage points over period
Waterbodies and marshland(incl. related water classes)	Marshland 410,796 ha + water bodies 408,835 ha + river 128,438 ha + river bank 897,906 ha (detailed 2016 legend); comparative Table 6 uses 1.85 million ha (2.9%) aggregated	n/a	1.2 million ha (1.8%)— Table 7 waterbodies & marshland 1,164,311 ha	Decrease (1.85 → 1.2 million ha)
Snow cover	497,236 ha (0.77%)	n/a	0.6 million ha (0.9%)— Table 7: 558,431 ha	Slight increase (0.5 → 0.6 million ha)
Total agricultural land(irrigated herbaceous + rainfed + fruit + vine, approx.)	~11–12% depending on whether marginal irrigated/karez classes are included	NSIA: total cultivated/arable often reported as >3 million ha (irrigated ~2.1 million ha; rainfed ~0.9 million ha) — current season cultivation, not full LC agricultural mask	11% (7.1 million ha) in abstract(5.4% irrigated herbaceous + 4.8% rainfed + orchards/vineyards remainder)	Agricultural composition shifts: total irrigated share up slightly; rainfed down; permanent crops up

4.1 Interpretation of the major changes (2016 → 2025 FAO)

- Irrigated systems: overall irrigated land (herbaceous + orchards + vineyards) rose from 5.9% to 6.2%, but irrigated herbaceous cropland itself declined slightly (–115,923 ha). Growth is concentrated in permanent irrigated crops (fruit trees especially).
- Rainfed systems: sharp reduction (–656,566 ha), attributed in the Atlas to climatic stress (droughts/floods) and the shift from multi-year potential mapping to single-year observation.
- Rangeland loss and bare/sand expansion: rangeland –1.7 million ha; bare areas and sand cover increase — interpreted as land degradation / desertification pressure.
- Urbanization: built-up area roughly doubled (0.5% → 1.0%), notably around Kabul, Herat and Kandahar.
- Water/marsh decline: 1.85 → 1.2 million ha, highlighting hydrological stress.

- Method caveat (critical): the Atlas itself warns that 2016 and 2024 methodologies, sensors and processing differ substantially; part of apparent agricultural/rangeland reduction reflects transition from aggregated multi-year/potential estimates to a single-year observed map.

4.2 How to read NSIA figures against FAO atlas figures

NSIA's subsequent updates are valuable for seasonal cultivated area and crop production, but they are not direct substitutes for FAO land-cover class areas. In particular:

- NSIA cultivated land (~3+ million ha) is typically "land under cultivation" in a season, whereas FAO rainfed + irrigated LC classes include the mapped agricultural land-cover mask (active and/or classified agricultural surfaces) and are therefore often larger.
- NSIA irrigated/rainfed splits are useful for food-security monitoring but use operational definitions that may exclude orchards/vineyards or fallow/marginal classes treated differently in LCCS/LCML.
- For policy comparison of land-cover change across the full landscape (rangeland, bare, forest, built-up, water), the FAO 2016 vs 2025 atlas comparison (Table 6) is the appropriate like-for-like reference—subject to the methodological caveat above.

5. Primary sources consulted

- FAO. 2025. Land Cover Atlas of Afghanistan 2024. Rome. <https://doi.org/10.4060/cd3186en> (uploaded PDF: LAND_COVER_ATLAS_OF_AFGHANISTAN_2025_V15B).
- FAO. 2016. The Islamic Republic of Afghanistan Land Cover Atlas. Rome. <https://openknowledge.fao.org/handle/20.500.14283/i5457e> and related PDF i5043e.
- FAO geospatial catalogue: Aggregated Land Cover Database of the Islamic Republic of Afghanistan (2010) — SPOT-4/Landsat, LCCS, MADCAT, object-based mapping metadata.
- UN Geospatial / UN Cartographic map practices and standard boundary disclaimers; FAO Data Lab map disclaimer pages (including contested-boundary language).
- NSIA public communications on satellite-based cultivated-area / wheat-area assessments (post-2016 operational updates).

This briefing note is an analytical synthesis for reference use. Boundary representations follow the Atlas/UN disclaimer and do not imply determination of sovereignty or frontiers.